

When “Unpopular Omiyage” Breaks the Experiment: A Failed Test of Informativity and Memory Robustness in Japanese

Jeremy Lim (281769)

June 23, 2026

Abstract

This study adapts the strategic-memory-allocation framework of Xu and Futrell (2025) to a non-local negative-polarity-item (NPI) dependency in Japanese. In a 2×2 within-participants acceptability judgment task, I crossed DEPENDENCY LENGTH (Short vs. Long) with SURPRISAL (Low vs. High, manipulated through the typicality of a pre-nominal modifier, for example “popular” vs. “unpopular” souvenir), and measured the acceptance rate (AR) and the judgment response time (RT). The first hypothesis was supported: short dependencies were more acceptable and faster to judge than long ones. The key prediction, a Dependency Length by Surprisal interaction, was not supported in either measure. Acceptance was low overall, and high-surprisal sentences were accepted much less often even though all targets were grammatical. I argue that the experiment largely failed to test the hypothesis for three mutually reinforcing reasons: the high-surprisal targets were themselves semantically odd, so interpreting them added processing cost; a more cautious response style, plausibly tied to language and testing differences, depressed acceptance and slowed responses; and the instructions did not clearly separate truly ungrammatical sentences from sentences that are merely unusual but acceptable. Because Xu and Futrell’s effect appeared only in online reading times, the offline null reported here does not by itself argue against strategic memory allocation. I conclude with concrete design changes for a more sensitive replication.

1 Background

Next-word prediction, the foundation of large language models (LLMs), is driving the rapid development of artificial intelligence. Human language comprehension has likewise been modeled as expectation-based: more predictable linguistic units are easier to process (Levy, 2008). Predictability also has a memory side. Because a predictable unit can be reconstructed from a lossy memory of its context (Futrell, Gibson, & Levy, 2020), it may receive less careful encoding; an unpredictable but informative unit cannot be guessed back in the same way, so it may require a more robust representation. This raises the question of how units that are hard to predict but rich in information are processed, and remembered, differently.

Xu and Futrell (2025) propose **strategic memory allocation**: because working memory is limited and noisy, comprehenders rationally devote more encoding effort to less predictable, more informative units, building representations that are more robust against interference. They test this through the agreement-attraction effect in self-paced reading. In their Experiment 1 they manipulate the surprisal of a singular target noun (e.g., *the evil monster* vs. *the cute monster*) and find that more surprising target nouns show a *weaker* agreement attraction effect. They interpret this as evidence that informative units receive more robust memory representations and are therefore less susceptible to distractor interference at retrieval.

This study borrows two ideas from this work. First, I treat surprisal (informativity) as a property of a critical noun phrase. Second, building on the same strategic-memory-allocation idea as applied to dependency length, I additionally manipulate the *distance* between the critical phrase and its licenser. If informative units are stored more robustly, higher surprisal should help a long-distance dependency survive in memory, predicting an interaction between surprisal and dependency length that would surface as shorter response times in the long, high-surprisal condition.

I chose Japanese as the test language for two reasons. First, Japanese is an SOV, head-final language and may pattern differently from English, so the same syntactic manipulation need not give the same result across languages; testing it therefore contributes to the study of universal grammar, linguistic typology, and cross-linguistic research. Second, its head-final structure and overt negative-polarity-item (NPI) marking make non-local dependencies easy to construct, which allows the predicted surprisal-by-dependency-length interaction to be observed.

2 Research Design

Research question and hypotheses. Does the informativity (surprisal) of a critical NP modulate the cost of a long-distance dependency in Japanese? Following strategic memory allocation, I predicted: (H1) shorter dependencies are easier (higher acceptance, faster RT) than longer ones; (H2) the cost of a long dependency is *reduced* when the critical NP is high in surprisal, that is, a $\text{DEPENDENCY LENGTH} \times \text{SURPRISAL}$ interaction. In detail, this interaction predicts that going from short to long should raise RT sharply for low-surprisal items, but should raise RT only slightly for high-surprisal items.

Design and variables. I used a 2×2 within-participants design crossing two manipulated factors: DEPENDENCY LENGTH (*Short* vs. *Long*: the linear distance between the critical NPI phrase *N-shika* ‘only N’ and the clause-final negated verb licensing it) and SURPRISAL (*Low* typical vs. *High* atypical: manipulated through a pre-nominal modifier making the critical NP more or less expected). The four conditions are coded ea (Short/Low), eb (Long/Low), ec (Short/High), and ed (Long/High).

Adapting the framework to Japanese. Because Japanese lacks subject–verb number agreement, I did not replicate agreement attraction directly. Instead I instantiated a non-local dependency with the negative-polarity item *shika* ‘only’, which obligatorily requires clause-final negation as its license. This dependency must be held in memory across intervening material, paralleling the target-noun–verb dependency in Xu and Futrell (2025). A fully glossed sample item is given in (1); the critical *shika...neg* dependency is what must be maintained.

- (1) Kankokyaku-wa gaidobukku-o shinkenni yonda-atode ninki-ga-arū omiyage-shika sagas-anakatta.
 tourist-TOP guidebook-ACC seriously read-after popular souvenir-ONLY search-NEG.PST
 ‘The tourist searched only for popular souvenirs after seriously reading the guidebook.’ (condition ea: Short/Low)

Dependency length was manipulated by placing the adverbial subordinate clause (*after seriously reading the guidebook*) either before the *shika*-NP (Short) or between it and the verb (Long). Surprisal was manipulated by the modifier on the *shika*-NP (*ninki-ga-arū* ‘popular’, Low vs. *ninki-ga-nai* ‘unpopular’, High). The four surface strings appear in Table 1.

Table 1: Sample target item in the four conditions (romanized). The critical *shika*–negation dependency spans from the marked NP to the final negated verb.

Condition	Sentence
ea Short/Low	Kankokyaku-wa gaidobukku-o shinkenni yonda-atode ninki-ga-arū omiyage-shika sagas-anakatta.
eb Long/Low	Kankokyaku-wa ninki-ga-arū omiyage-shika gaidobukku-o shinkenni yonda-atode sagas-anakatta.
ec Short/High	Kankokyaku-wa gaidobukku-o shinkenni yonda-atode ninki-ga-nai omiyage-shika sagas-anakatta.
ed Long/High	Kankokyaku-wa ninki-ga-nai omiyage-shika gaidobukku-o shinkenni yonda-atode sagas-anakatta.

Stimuli and fillers. There were 20 target items, each shown in one condition per participant under a Latin-square distribution, plus 40 fillers (12 designed to be acceptable and 28 containing clear and diverse grammatical violations) and 4 practice items. The fillers were included mainly to disguise the purpose of the experiment and to discourage strategic responding, so that participants could not infer the design and answer by rule rather than by intuition; the contrast between acceptable fillers (expected “F Yes”) and clearly ungrammatical fillers (expected “J No”) also gives a rough check that participants were attending to the task.

3 Method

Participants. Twenty-six people took part ($N = 26$). All were native speakers of Japanese; most also reported some experience with a second language (L2), though a few reported none, so the sample is largely Japanese-dominant

rather than uniformly monolingual.¹ The experiment link was sent to friends of the author by email. Before starting, each participant gave informed consent and then completed a short background questionnaire that asked for birth year, gender (female, male, or other / prefer not to answer), an email address, any non-native language used fluently or in daily life, and the prefectures or countries where they had lived together with the corresponding ages.

Materials. The materials are described above in the Research Design section and were drawn from the class stimulus set. The target sentences were created by the LNG 223 course participants, checked by Claude (AI) for lexical overlap across items, and revised again in class; the fillers were generated by Claude (AI) and then checked by native Japanese speakers in the course. In brief, there were 20 target item sets. Each set was written in all four conditions of the 2×2 design (Dependency Length: Short vs. Long $\times$ Surprisal: Low vs. High), giving $20 \times 4 = 80$ target sentences in total, that is, 20 sentences per condition. There were also 40 filler sentences (12 acceptable and 28 unacceptable, spread over six grammatical types) and 4 practice sentences. The 80 target sentences were divided into four lists with a Latin square, so that each participant saw each of the 20 target items exactly once and in only one condition; across the four lists, every item appeared in every condition. PCIBex assigned each participant to one of the four lists, so the lists were of approximately equal size. Each participant therefore judged 20 target trials and 40 fillers (60 experimental trials), preceded by 4 practice trials.

Rating scale. The judgment was a binary, two-alternative forced choice. For each sentence the participant decided whether it was “natural” or “unnatural” and answered with one of two keys: the **F** key for natural and the **J** key for unnatural. Participants were asked to rest both index fingers on these keys and to respond as quickly and as accurately as possible. The instructions stressed that they should answer by first impression, that the task was not a school grammar test, and that they should attend to whether the whole sentence reads smoothly rather than to single-character differences.

Procedure. The experiment was built in PCIBex, a JavaScript-based platform for web-based experiments (Zehr & Schwarz, 2018), and was run online in each participant’s own web browser, with no restriction on the device used. After the consent form and the background questionnaire, participants read the task instructions and worked through four examples. They then completed the practice trials and the main session, in which the target and filler sentences were presented one at a time in a self-paced stream. No breaks were allowed once the main session began, and a session lasted roughly 10 minutes.

Dependent measures. Two dependent variables were analyzed: the **Acceptance Rate (AR)**, the proportion of trials on which a sentence was judged natural (the **F** response), and the **Response Time (RT)**, the time from sentence onset to the key press, recorded in milliseconds.

4 Analysis

Software and packages. All data processing, statistical modeling, and visualization were carried out in Python 3.10, using `pandas` and `numpy` for data handling, `statsmodels` (Seabold & Perktold, 2010) for the statistical models, and `matplotlib` for the figures. The cleaning and aggregation pipeline reproduces the analysis flow taught in class by Prof. Hajime Ono, which was originally written in R with the `tidyverse` and `ggplot2` packages.

Outlier and incomplete-data handling. Non-experimental rows and practice trials were removed first. For the acceptance-rate analysis, all target trials were kept and fillers were excluded. For the response-time analysis, only correct (accepted) target trials were used, and the response times were trimmed in two steps: an absolute cut that removed responses of 8000 ms or longer, followed by a per-condition cut that removed responses more than 2.5 standard deviations from the condition mean. Table 2 reports how many trials remained after each step. The one participant whose record could not be matched cleanly is handled as described in the Method footnote.

Statistical models. Both factors were effect-coded, so each main effect and the interaction can be read straight off the model. I analyzed each dependent variable with two models. Acceptance is a yes/no response, so I used a logistic model (a generalized estimating equation, GEE) that fits binary data and allows for the fact that each participant contributes many correlated trials. Response time is continuous, so I used a linear mixed-effects model with Dependency Length, Surprisal, and their interaction as fixed effects, plus random intercepts for participants and for items; the random intercepts let overall differences between people and between sentences be set aside so

¹Participants were keyed by the PCIBex submission timestamp, following the class analysis script. One IP-hash appears under two timestamps, so a stricter participant definition yields $N = 25$; this does not change the qualitative pattern.

they do not distort the effects of interest. For each measure I also ran a simple 2×2 repeated-measures ANOVA on the by-participant condition means as a cross-check. I chose the GEE and mixed-effects models because, unlike an ANOVA on averaged scores, they use all of the trial-level data and explicitly handle the repeated measurements from each participant and item.

Visualization. Condition means were drawn as a grouped bar chart for acceptance and as a point-and-line interaction plot for response time (Figure 1), with error bars showing one standard error. The model estimates were shown as a coefficient plot with 95% confidence intervals (Figure 2), where an interval that excludes zero marks a reliable effect. By-participant and by-item means were plotted as sorted bar charts (Figure 3) to check that no single participant or item drove the pattern, and the participant-level link between acceptance and speed was shown as a scatter plot with a fitted line (Figure 4), and per-participant response times were plotted as one small panel per participant (Figure 5) to show how complete each participant’s data were and whether individual patterns matched the predicted interaction. The exploratory inclusive analysis is shown in three further figures: Figure 6 compares the condition-mean response times under the correct-only and the inclusive standards; Figure 7 shows, for each participant, how many of the four conditions had usable data under each standard; and Figure 8 repeats the per-participant panels under the inclusive standard.

Table 2: Trial counts through the response-time data-cleaning pipeline.

Step	Trials
Correct target trials with a valid RT	243
After absolute cut (≥ 8000 ms removed)	178
removed by absolute cut	65
After 2.5 SD per-condition cut	176
removed by SD cut	2

5 Results

Tables and figures in this section are ordered from the most relevant to the hypothesis to exploratory robustness checks.

Condition means. Condition means for both dependent variables are in Table 3 and Figure 1. Acceptance was much higher for short than long dependencies (about .65 vs. .28) and slightly higher for low- than high-surprisal items; response times were faster for short dependencies (reliable in the mixed-effects model, though not in the underpowered by-participant ANOVA; see below).

Table 3: Condition means for Acceptance Rate (AR) and Response Time (RT, ms). SD and SE are computed across participant means (two-stage aggregation).

DepLength	Surprisal	Acceptance Rate			Response Time (ms)		
		Mean	SD	SE	Mean	SD	SE
Long	High	0.235	0.286	0.056	5152	2144	429
Long	Low	0.316	0.333	0.065	5396	2184	437
Short	High	0.610	0.300	0.059	4991	1517	303
Short	Low	0.679	0.250	0.049	5203	1437	287

Hypothesis tests. For acceptance, the repeated-measures ANOVA showed a large effect of Dependency Length, $F(1, 25) = 42.32, p < .001$, a small effect of Surprisal that fell just short of significance, $F(1, 25) = 4.02, p = .056$, and no interaction, $F(1, 25) = 0.02, p = .878$. The logistic GEE (Table 4) agreed: acceptance was higher for short dependencies ($\beta = 1.55$ log-odds, $p < .001$) and lower for high-surprisal items ($\beta = -0.36, p = .034$), with no reliable interaction ($\beta = 0.13, p = .69$). For response time, the by-participant repeated-measures ANOVA found no reliable effect of Dependency Length, $F(1, 7) = 2.67, p = .15$, but only eight participants had data in all four cells,

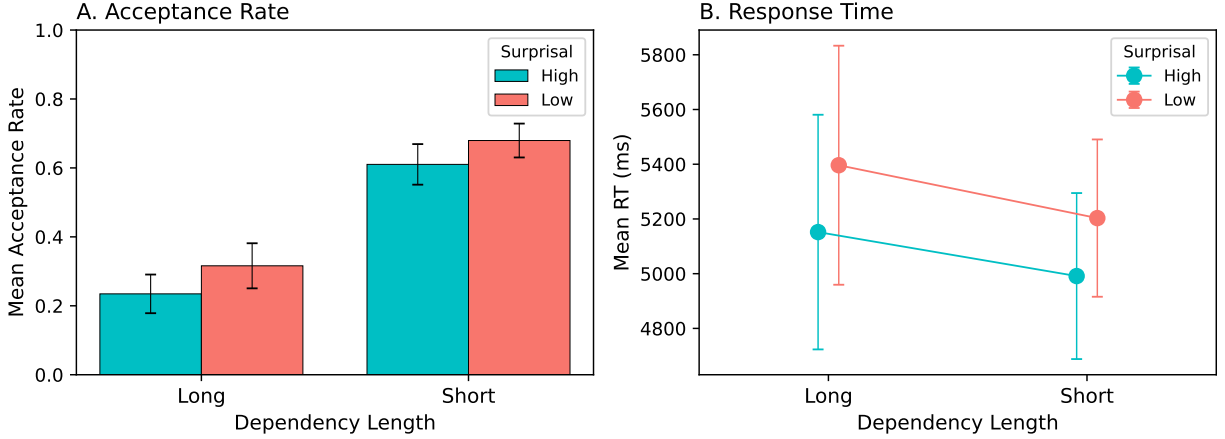


Figure 1: Condition means by Dependency Length and Surprisal. **A.** Mean Acceptance Rate (bars). **B.** Mean Response Time in ms (points). Error bars are ± 1 SE computed across participant means.

so this test is badly underpowered. The trial-level linear mixed-effects model (Table 5), which uses all usable trials, did show a reliable effect of Dependency Length ($\beta = -832$ ms, $p = .001$); Surprisal ($\beta = +398$ ms, $p = .077$) and the interaction ($\beta = -125$ ms, $p = .78$) were not reliable. Thus **H1 (shorter dependencies easier) was supported in both measures, while H2 (the predicted Dependency Length $\times$ Surprisal interaction) was not supported in either measure.** Figure 2 shows the coefficient estimates with 95% confidence intervals: in both panels the Dependency Length interval lies clearly on one side of zero (the H1 effect), the Surprisal interval sits near zero, and the Dependency Length $\times$ Surprisal interval overlaps zero (no H2 interaction).

Table 4: Logistic GEE on acceptance (effect-coded predictors, clustered by participant). Estimates are in log-odds.

Predictor	Estimate	SE	95% CI	z	p
Intercept	-0.176	0.198	[-0.563, 0.212]	-0.89	0.374
DepLength (Short vs Long)	1.554	0.271	[1.023, 2.084]	5.74	<.001
Surprisal (High vs Low)	-0.357	0.168	[-0.687, -0.027]	-2.12	0.034
DepLength $\times$ Surprisal	0.130	0.330	[-0.515, 0.776]	0.40	0.692

Table 5: Linear mixed-effects model on response time (ms) with crossed random intercepts for participant and item; effect-coded predictors.

Predictor	Estimate	SE	95% CI	z	p
Intercept	5408.5	320.0	[4781.3, 6035.6]	16.90	<.001
DepLength (Short vs Long)	-832.3	254.1	[-1330.4, -334.3]	-3.28	0.001
Surprisal (High vs Low)	398.3	225.2	[-43.2, 839.7]	1.77	0.077
DepLength $\times$ Surprisal	-125.4	456.6	[-1020.3, 769.6]	-0.27	0.784

By-participant and by-item variation. Acceptance and response time varied widely across participants and items, but no single participant or item drives the results (Figure 3). Across participants, acceptance ranged from about .10 to .85, and across items from about .12 to .69. The grand means (dashed lines) are about .47 for acceptance and about 4,980 ms for response time, and the sorted bars rise smoothly rather than jumping at one or two cases, so the variation is gradual rather than driven by a few outlying participants or items.

Acceptance-RT correlation. Across participants, mean Acceptance Rate and mean Response Time showed a weak negative correlation that was not statistically significant (Pearson $r = -0.16$, 95% CI $[-0.52, 0.25]$, $p = .45$; Spearman $\rho = -0.20$, $p = .33$; $n = 25$, the 25 of 26 participants with usable trimmed RT; Figure 4, Table 6). As

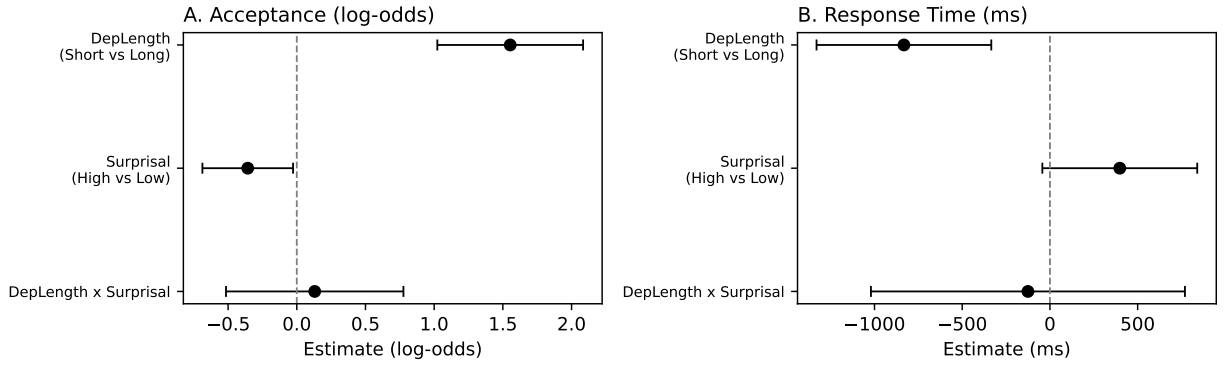


Figure 2: Model coefficient estimates with 95% confidence intervals. **A.** Logistic GEE on acceptance (log-odds). **B.** Linear mixed-effects model on response time (ms). The dashed line marks zero; an interval excluding zero indicates a reliable effect.

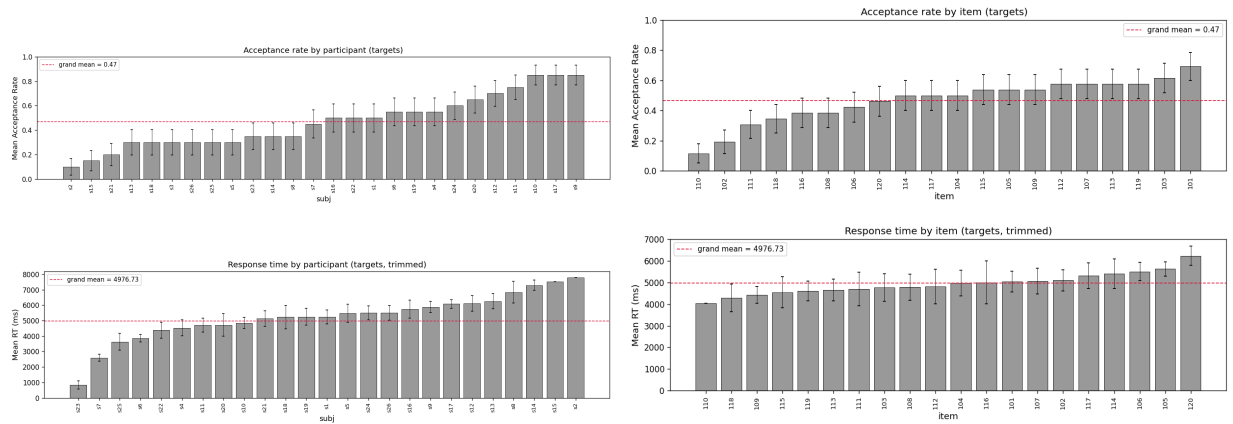


Figure 3: Individual-level means (targets only), sorted, with the grand mean (dashed line). Top: Acceptance Rate by participant (left) and item (right). Bottom: Response Time by participant (left) and item (right). Error bars are ± 1 SE.

Figure 4 shows, several participants cluster in the low-acceptance, long-RT region, in line with a slow-and-cautious response tendency.

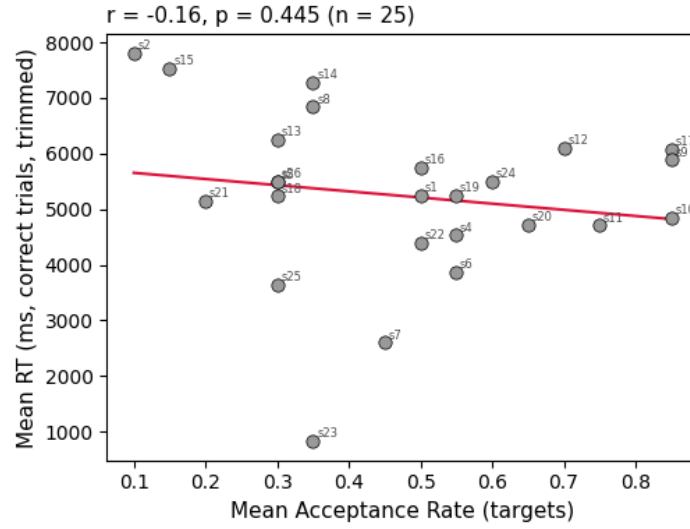


Figure 4: Participant-level relationship between mean Acceptance Rate (targets) and mean Response Time (correct, trimmed trials). Each point is one participant; the line is an OLS fit.

Table 6: Participant-level correlation between mean Acceptance Rate and mean Response Time (correct, trimmed trials). 95% CI shown for the Pearson coefficient (Fisher z).

Method	Coefficient	95% CI	p	n
Pearson	-0.160	[-0.522, 0.251]	0.445	25
Spearman	-0.202	–	0.332	25

Per-participant condition coverage (correct-only). Figure 5 plots each participant’s mean RT by condition using the correct-only trials. Many panels are incomplete: a large number of participants have data in only one or two of the four conditions, because most of their trials were either rejected or removed by the response-time cuts. This sparse coverage is what motivates the exploratory inclusive analysis that follows.

Exploratory: inclusive RT analysis. Because the correct-only RT data are sparse (176 observations; only 8 participants with all four cells), I re-ran the identical trimming pipeline on all target trials, accepted and rejected alike. This more than doubled the usable data to 386 observations (Table 7) and brought most participants to full condition coverage (Figure 7; per-participant profiles in Figure 8). At the group level, however, the inclusive standard lowered overall RTs and numerically reversed the Dependency Length pattern relative to the correct-only analysis (Figure 6, Table 8).

Table 7: Usable response-time observations under the main (correct-only) standard versus the inclusive standard that also keeps trials judged unacceptable. Both apply the same 8000 ms absolute cut and 2.5 SD per-condition cut.

	Correct-only (main)	Inclusive (all trials)
Trials entering trimming	243	520
Removed by absolute cut	65	117
Removed by SD cut	2	17
Final RT observations	176	386

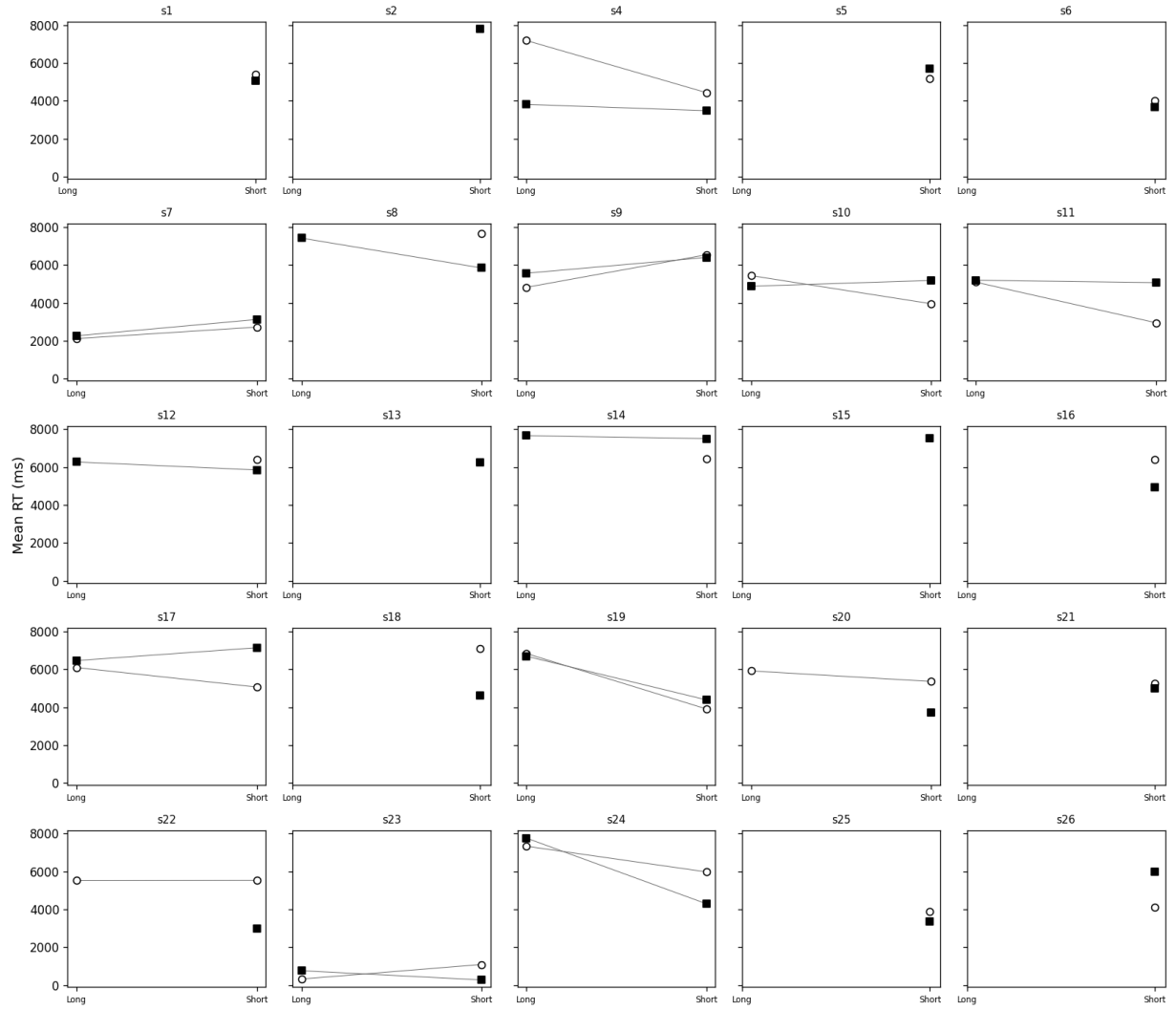


Figure 5: Per-participant mean RT by Dependency Length (x-axis) and Surprisal (open circle = High, filled square = Low), one panel per participant, using the correct-only trials (after the 8000 ms cut). Incomplete panels show how few usable response times remain per participant under the correct-only standard.

Table 8: Condition mean RT (ms) under the correct-only standard versus the inclusive standard (all target trials). SE across participant means.

DepLength	Surprisal	Correct-only		Inclusive	
		Mean	SE	Mean	SE
Long	High	5152	429	4522	220
Long	Low	5396	437	4641	271
Short	High	4991	303	4706	243
Short	Low	5203	287	4800	218

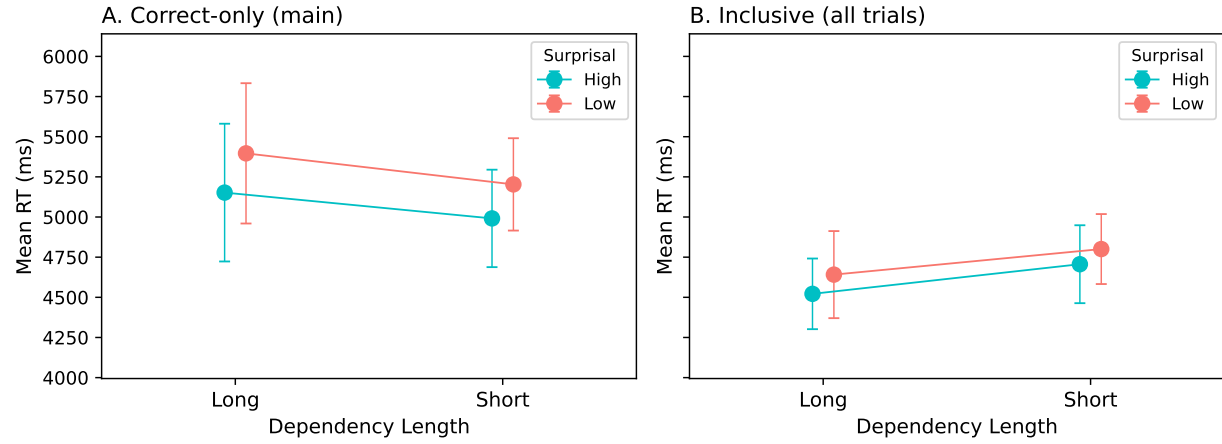


Figure 6: Condition-mean Response Time by Dependency Length and Surprisal. **A.** Correct-only standard (main analysis). **B.** Inclusive standard (all target trials). Error bars are ± 1 SE across participant means; both panels share the same y-axis.

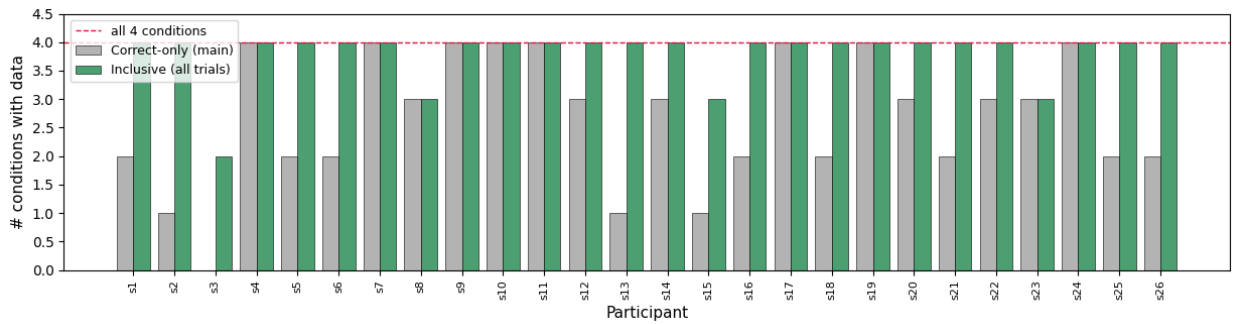


Figure 7: Number of design cells (out of four) with usable RT data per participant, under the correct-only standard (grey) versus the inclusive standard (green). The inclusive standard brings most participants to full coverage (dashed line).

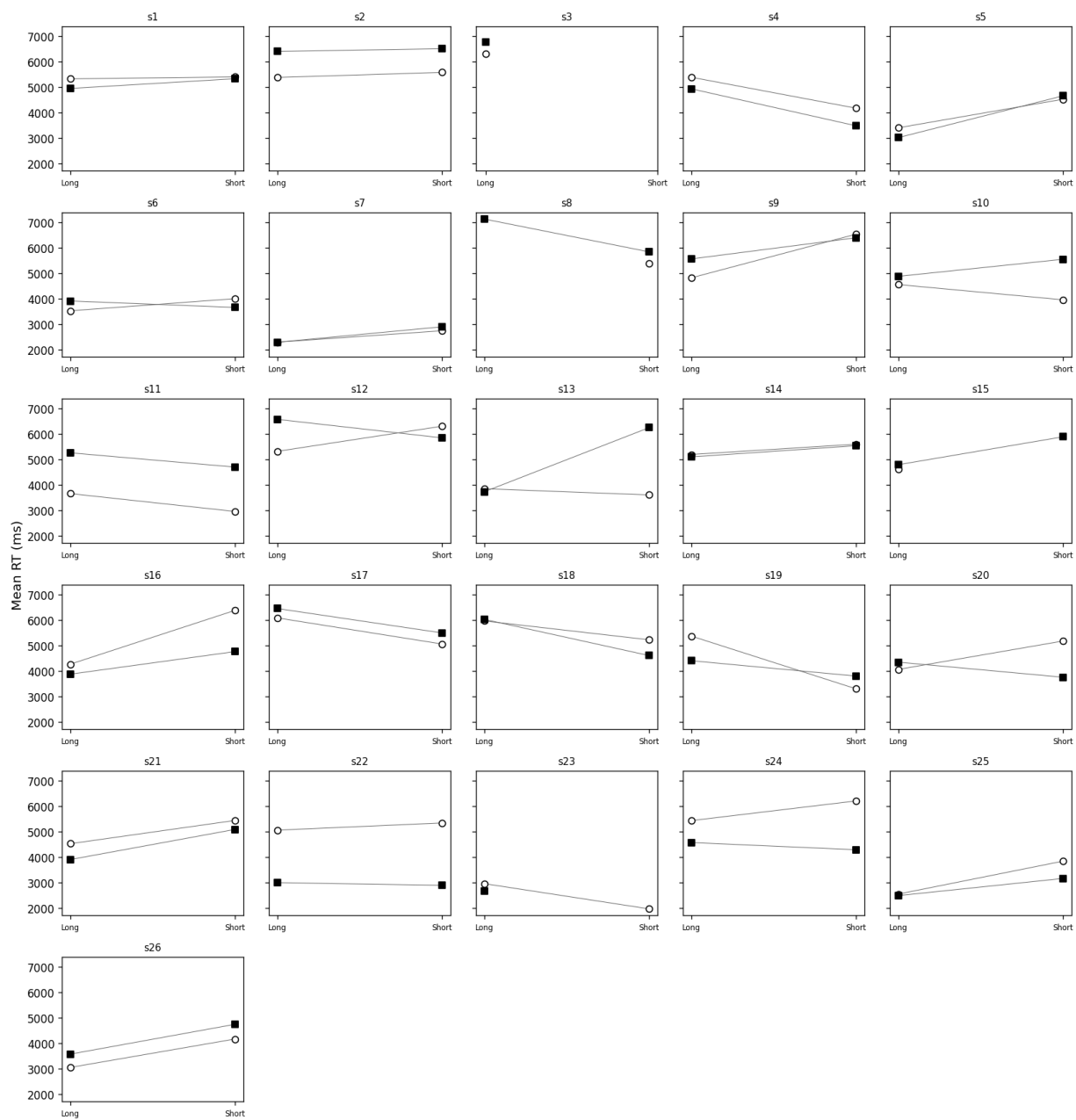


Figure 8: Per-participant mean RT by Dependency Length (x-axis) and Surprisal (open circle = High, filled square = Low) under the inclusive standard. Far more participants now show a complete two-line profile than under the correct-only standard.

6 Discussion

Were the hypotheses supported?

The results clearly supported the first hypothesis but not the second. Short dependencies were easier than long dependencies in both measures. Acceptance was much higher for short than for long dependencies (Table 3, Figure 1), and short dependencies were read faster (Table 5). This supports H1. The main prediction of strategic memory allocation, however, was that high surprisal should reduce the cost of a long dependency and produce a Dependency Length by Surprisal interaction (H2). This interaction did not appear in acceptance or in response time. On the surface, then, the experiment did not support the hypothesis. In my view, three mutually reinforcing problems with the task and materials make this result hard to read as real evidence against the hypothesis.

Why the second hypothesis was not supported

First, the target sentences themselves may have been too unusual in meaning. To manipulate surprisal, the high-surprisal items combined a grammatical but uncommon modifier with the rest of the sentence, for example searching only for an “unpopular” souvenir, which describes a situation that Japanese speakers rarely meet in daily life. Such sentences are grammatical, but they are semantically odd and therefore take extra time to interpret, because the reader must work out a strange meaning. Several high-surprisal items show this clearly. For example, *Gaka-wa shiroi karasu-shika ...egak-anakatta* ‘the painter painted only white crows ...’ was accepted on only 12% of trials with a mean response time of about 7300 ms, and *Jukensei-wa kantan-na shiken-shika ...uke-nakatta* ‘the student took only easy exams after cramming the reference book’ on 12% with about 6650 ms; one item (‘only un-soothing scenery spread before the hiker’) was accepted by no participant at all. These sentences are grammatical, but their unusual meanings force extra interpretive work, so their very low acceptance and long response times reflect semantic difficulty rather than the memory mechanism under study. As a result, the offline measures reflect not only the effect I wanted to observe, the interaction of surprisal and dependency length, but also the extra cost of making sense of grammatical yet unusual expressions. Because these two sources of difficulty are mixed together, the offline result is hard to interpret cleanly as a test of the hypothesis. This points to a deeper design tension: high surprisal is, by definition, semantically unpredictable and informative, which is exactly the property the experiment needs, but if a phrase is pushed so far that it tips into sounding unnatural, participants reject the sentence instead of processing it. A future version therefore has to manage the boundary between “high surprisal” and “unnatural” carefully, so that the critical phrase still draws robust memory and attention while the sentence as a whole is still judged natural; achieving this calls for careful manipulation and norming of the high-surprisal phrase.

Second, the response style of the participants may also have lowered acceptance. The participants were Japanese university students, and Japanese respondents are often careful and conservative in judgment tasks. This cautiousness may itself be a cultural factor: in many testing situations Japanese respondents tend to avoid committing to a confident “yes” when they are unsure, and instead take more time and choose the safer “unnatural” answer. Several features of the data fit this idea. Response times were very long, with condition means around 5000 ms, and many trials were slower than the 8000 ms cut and were removed (Table 2). Acceptance was low even for sentences that were meant to be acceptable, and the differences between participants were large (Figure 3). The participant-level relationship between the two measures points the same way: the negative correlation between acceptance and response time (Pearson $r = -0.16$; Figure 4) means that participants who accepted fewer sentences also tended to be slower, which is just what we would expect if cautious participants both hesitate longer and lean toward rejecting. The correlation is weak and not statistically reliable, so I treat it only as suggestive, but its direction is consistent with the response-style account. A cautious style would affect the borderline high surprisal items the most, which would both create the surprisal main effect and hide any helpful interaction. If this is correct, the acceptance rate reflects a response bias as much as it reflects the underlying representation.

Third, I think the instructions given before the experiment were not clear enough. All target sentences were designed to be grammatical and acceptable, so participants should have accepted them most of the time. Instead, overall target acceptance was only about .47, and even the easiest condition (short and low surprisal) reached just .68 (Table 3). High surprisal sentences were accepted much less often, and surprisal reliably lowered acceptance (Table 4). The on-screen instructions did ask participants to decide whether each sentence was “natural” or “unnatural”, to answer by first impression, and to treat the task as everyday language use rather than a school grammar test,

and four worked examples were shown. Although the instructions did include examples of natural sentences, the number of examples and practice trials was too small to shift participants' mindset, so many of them kept treating an unusual but grammatical sentence as unnatural. The high-surprisal targets were built to be exactly this kind of item, for example with the atypical modifier *ninki-ga-nai* 'unpopular', and participants appear to have rejected them for being uncommon rather than ungrammatical. If "natural" conflates grammaticality with typicality, the acceptance rate no longer measures the strength of the memory representation, which is what the hypothesis is about. A possible direction of future revision would be to add many more practice trials before the main session, in particular grammatical-but-unusual items with corrective feedback, so that participants build the correct mindset and learn that an unusual sentence can still count as "natural" before the real items begin.

Relation to Xu and Futrell (2025)

All three problems concern the offline measure, which shapes how the results relate to Xu and Futrell (2025). In their study, the critical effect (target-noun surprisal modulating the agreement-attraction effect) was reliable in the online, region-by-region reading times but not in the offline acceptability judgments. The present design had no online measure, only a single whole-sentence judgment time, so even a real effect would have been hard to detect: the clear locality effect (H1) shows that the method can catch a large processing cost, but not the smaller surprisal modulation. The missing interaction is therefore consistent with their offline null and does not by itself argue against strategic memory allocation.

The inclusive response-time analysis

The exploratory inclusive analysis (which also kept the rejected trials) shows why the response-time data must be treated with care. Although it more than doubled the usable data and gave most participants full condition coverage (Table 7, Figures 7 and 8), the group pattern changed: overall RTs were lower and the Dependency Length effect reversed direction relative to the correct-only analysis (Figure 6, Table 8). Because rejected trials may reflect guessing or off-task behavior that itself depends on the condition, this larger dataset is not necessarily more trustworthy. Rather than giving a clearer answer, the reversal shows that the direction of the response-time effect depends heavily on which trials are included, so the inclusive result should be read with caution.

A closer look at the individual panels reinforces this caution. In the correct-only per-participant plot (Figure 5), s4 most clearly shows the pattern predicted by H2, in which high surprisal lowers the cost of the long dependency, although the sparse per-participant data make this reading tentative. This participant reported no fluent second language and has lived in Yamanashi Prefecture from birth to the present, making s4 the most clearly monolingual participant in the sample. That the most H2-consistent participant is also the most clearly monolingual one suggests that second-language experience may matter, and that a sample of strictly monolingual native speakers could give a different result. Under the inclusive standard (Figure 8), by contrast, not a single participant shows the predicted pattern. The inclusive analysis therefore adds data but no support for the hypothesis, which confirms that, for the present purpose, including the rejected trials is uninformative.

Limitations

This study has four main limitations. First, the instructions and practice were not enough to teach participants the difference between unnatural and merely unusual sentences. Second, although the stimuli were piloted, the pilot results were not reflected carefully enough in revising the items, so some high-surprisal items remained too odd. Third, the design did not take cultural differences in response style into account. Fourth, the usable response-time sample was small.

Future directions

Beyond the fixes noted above (clearer instructions with practice and feedback, and stimuli normed to separate surprisal from plausibility), future versions should add an online measure such as self-paced reading or eye tracking, replace the binary judgment with a graded 0–100 scale, test more and ideally strictly monolingual participants, and compare response criteria across languages to test the cautious-responding account.

References

- Futrell, R., Gibson, E., & Levy, R. P. (2020). Lossy-context surprisal: An information-theoretic model of memory effects in sentence processing. *Cognitive Science*, 44(3), Article e12814.
- Levy, R. (2008). Expectation-based syntactic comprehension. *Cognition*, 106(3), 1126–1177.
- Seabold, S., & Perktold, J. (2010). statsmodels: Econometric and statistical modeling with Python. In *Proceedings of the 9th Python in Science Conference* (pp. 92–96).
- Xu, W., & Futrell, R. (2025). Informativity enhances memory robustness against interference in sentence comprehension. *Journal of Memory and Language*, 142, Article 104603.
- Zehr, J., & Schwarz, F. (2018). *PennController for Internet Based Experiments (IBEX)*. <https://doi.org/10.17605/OSF.IO/MD832>

AI Usage Record

- **Tool:** Claude (Anthropic), used as a coding and writing assistant.
- **Stimulus creation:** The target sentences were written by the LNG 223 course participants and checked by AI for lexical overlap across items; the filler sentences were generated by AI and then checked by native Japanese speakers in the course.
- **Data analysis code:** AI wrote the Python pipeline that cleans the PCIBex export, computes acceptance/RT condition means, performs RT trimming, and fits the ANOVA, logistic GEE, and mixed-effects models.
- **Figures and tables:** AI generated the figures and the LaTeX tables from the analysis output with captions.
- **Results and objective content:** Values reported in the Results section, and any other objective content (e.g., use of Python/data-processing libraries for corpus handling), were drafted by AI and checked by the author against the source data.
- **Discussion and analysis:** The Discussion and any other content involving subjective analysis or theoretical argumentation was written by the author, then revised by AI for clarity and phrasing; the author accepted, rejected, or further revised each suggested change.
- **Writing and tone:** AI reviewed the paper throughout for grammar and unified the academic tone across the whole article, proposing phrasing edits that the author then accepted, rejected, or revised.
- **Revision suggestions:** AI offered revision suggestions throughout, and the author adopted some of them and declined or rewrote others.
- **Verification:** The author checked the reported numbers against the corpus and analysis output, and confirmed all figures and tables match the data.